

JESUS F. SANCHEZ

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EDUCATION

Master of Science - University of Southern California (Los Angeles, CA) Expected May 2026
Aerospace Engineering, Concentration: Propulsion (**GPA: 4.0**)
Bachelor of Science - California State University, Northridge (Northridge, CA) August 2024
Mechanical Engineering, Concentration: Aerospace (**GPA: 3.7**)

WORK EXPERIENCE

NASA Jet Propulsion Laboratory, *Mechanical Engineering Intern* May 2022 - February 2024

- Led the design of complex electromechanical systems from design to manufacturing for Mars' Sample Retrieval Lander (SRL), focusing on design for manufacturability and assembly (DFM & DFA)
- Generated CAD models, engineering drawings using GD&T through Siemens NX and produced written instructions for build, assemble, and test
- Collaborated closely with technicians and the manufacturing team to support fabrication and troubleshoot hardware assembly issues, resulting in accurate and timely delivery of hardware

ENGINEERING PROJECTS & RESEARCH

Liquid Propulsion Laboratory (LPL), *Propulsion Engineer* September 2024 - Present

- Designed and tested a regeneratively cooled 3 kN Jet-A/LOX rocket engine focusing on thrust chamber geometry, nozzle contouring, and injector design; achieved 95% c^* efficiency in successful hot-fire testing
- Developed a bolt analysis tool to determine optimal bolt count, size, and torque to maintain a sealed chamber-injector interface under operational loads
- Conducted FEA of the thrust chamber assembly in Ansys Mechanical to evaluate stresses and seal integrity, leading to design revisions that minimized interface gap formation
- Performed CFD simulations to optimize injector geometry by reducing pressure losses and improving flow distribution
- Developed MATLAB-based thermal models to estimate maximum allowable firing duration and back-calculate combustion gas temperatures from thermocouple hot-fire testing data
- Conducted hot-fire data reviews to evaluate performance metrics and guided design updates, including optimized mixture ratio selection based on empirical results

Evaluation of Novel Scalable Swirl-Flame Hydrogen Combustor, *Research Fellow* May 2023 - August 2024

- Led the development and testing of a novel swirl-flame hydrogen combustor for gas turbine engines, achieving 60% hydrogen dilution; presented experimental results at the Western States Sec. Combustion Institute (WSSCI) conference
- Performed MATLAB-based data analysis and conducted CFD simulations in Ansys Fluent to evaluate flow fields and combustion behavior

ASME Human Powered Vehicle (HPV) Competition, *Fairing Team Member* August 2023 - May 2024

- Designed, analyzed (CFD), and fabricated a low-drag carbon fiber fairing with a drag coefficient of 0.43, a performance improvement over previous year's 0.50
- Performed FEA in SOLIDWORKS Simulation to evaluate and optimize fairing mounting arms, resulting in a 13% weight reduction and reduced stress concentrations
- Delivered Conceptual (CoDR), Preliminary (PDR), and Critical Design Reviews (CDR) to peers and industry professionals, demonstrating design decisions and testing results

NASA MINDS Competition, *Mechanical Design Sub-Team Member* September 2020 - May 2022

- **2022 CubeSat**: Conducted finite element analysis through SOLIDWORKS to optimize the design of a 4-inch cube satellite structure for weight reduction and material selection
- **2021 Autonomous Robot**: Designed the drivetrain system and analyzed (FEA) the chassis of a robot intended for nondestructive testing of space vehicles, enabling it to traverse 90-degree surfaces, achieving 3rd place in competition

SKILLS

Software: CAD (Siemens NX, SOLIDWORKS), CFD (Ansys Fluent, OpenFOAM), FEA (Ansys Mechanical, SOLIDWORKS Simulation), Cantera, LabVIEW, MATLAB, Python, Simulink, Teamcenter

Technical: Geometric Dimensioning and Tolerancing (GD&T), Tolerance Analysis, 3D Printing

COURSEWORK

Spacecraft Design, Compressible Gas Dynamics, Intro to Computational Fluid Dynamics, Fundamentals and Applications of Combustion, Partial Differential Equations, Rocket Propulsion, Aeropropulsion (see [more](#))